

Priyanshu Gupta

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EDUCATION

Indian Institute of Technology, Bombay (IIT Bombay)

Bachelor of Technology (Honors) in Electrical Engineering

(Nov'21 - April'25)

CPI:

SCHOLASTIC ACHIEVEMENTS

- Secured **Rank 25** at **ICPC Asia-West Championship**, representing IIT Bombay against top international teams (2025)
- Secured **Rank 24** (2024) and **Rank 69** (2023) at **India Regional Contests, Amritapuri, ICPC** (2023-24)
- Received **Best Project Award** (out of 60+ teams) in Electronics Design Lab for **Mech-Armed RC Rover** (2024)
- Secured **AIR 420 - JEE Advanced, AIR 493 - JEE Mains, AIR 1433 - KVPY** and **Rank 54 - WBJEE** (2021)
- Secured **Silver Medal** in **CPS Junior Maths Olympiad** in IX standard (2018)

PROFESSIONAL EXPERIENCE

Software Development Intern | BNY Mellon | Received **Full Time Offer**

(May'24-July'24)

- Worked on self-registration portal migration task; developed **API Service** in **SpringBoot** for user validation from database
- Implemented robust unit tests using **JUnit** and **Mockito** frameworks, ensuring reliability of **Spring MVC** applications
- Implemented **error & exception handling**; seamlessly deployed to **test environments** via **GitLab CI/CD** pipeline

Robotics Software Engineering Intern | Loopway Pvt Ltd

(May'23-July'23)

Part of the team developing autonomous-drive technology | Received **Letter of Recommendation**

- Researched and worked on **Inverse Measurement Probability Model** for the updation of Occupancy Grids
- Created a **Lane Marking Model** in **Python(ROS2)** to create rough lane boundaries, from sparse Lidar lane points
- Developed a real-time greedy algorithm **Waypoint Planner** for efficient path generation, prioritizing objective parameters

RESEARCH EXPERIENCE

Feature Selection Mechanisms for Multiomics Data | Prof. K. Jadhav, IITB

(Feb'24-April'24)

- Conducted a comparative analysis of **multi-omics** feature selection methods—**MRMR-mv**, **MAgentOmics**, **IVI**, **UMINT-FS**, and **ProMS**; benchmarked their performance on cancer datasets to optimize feature selection strategies
- Implemented **MRMR-mv** selection method; integrated **multi-omic** data; extracted features from ovarian cancer dataset

Coalitional Game Formulation in Multicast Networks | B.Tech Project

(Aug'24-April'25)

Guide: Prof. Gaurav Kasbekar, IITB

- Developed a **coalitional game theory framework** to model and optimize **multicast content distribution** in wireless networks
- Derived analytical conditions for **core emptiness** based on network parameters like data rate distributions and user counts

KEY PROJECTS

RL-Based Cache Replacement Policy | Course Project | Advanced Computer Architecture

(Nov'24)

- Trained a **Deep-Q Learning Agent** to analyse the dynamic data access patterns and captured relevant features
- Implemented Cache Replacement policy **RLR** with **Age Counters**, **Hit Registers**, and **Type Registers**, minimizing hardware overhead and leveraging model weight correlations to enhance **Last Level Cache (LLC)** efficiency across diverse benchmarks
- Evaluated **RLR** Policy using **ChampSim** Simulator, achieving **5% mIPC** speedup over Least Recently Used policy

Image Caption Generation | Course Project | Introduction to Machine Learning

(April'24)

- Designed an image captioning model using **pre-trained VGG16** as encoder and a network with **LSTM** layers as the decoder
- Fine-tuned the model on the **Flickr8K** dataset; achieved **BLEU-1** and **BLEU-2** scores of **0.57** and **0.34**, respectively

CPU Architecture Design (RISC & CISC) | Course Project | Microprocessors

(April'23)

- Designed an 8-register, 16-bit **Multistage Computing System** in **VHDL** for an assembly level **ISA** in **Intel Quartus**
- RISC**: Implemented **Hazard Control**, **Data forwarding** and **instruction decoders**; reduced stalling; achieved CPI of **1** in design
- CISC**: Modelled CPU as **FSM**. Designed the **hardware flowchart**, **datapath**, **control flow** and **decoding logic** for the given ISA

TECHNICAL SKILLS

Languages C, C++, Python, SQL, Java, VHDL, Assembly(x86), Embedded C, L^AT_EX

Softwares/Packages Git/GitHub, SpringBoot, Swagger, Jira, ROS2, ChampSim, Gem5, Valgrind, Quartus, Keil uVision5

Relevant Courses Adv. Computer Architecture • Intro. to Machine Learning • Adv. Probability • Linear Algebra • Intro. to Number Theory and Cryptography • Optimizations Course • Computer Programming

EXTRA CURRICULAR ACTIVITIES

- Codeforces rating: **1718 (Expert)** 🏆 | Codechef rating: **2077 (5-star)** 🏆 | LeetCode Solved: **250+** Problems 🏆 (Present)
- Finalists** of National Level competition **Robofest 3.0** in **Autonomous ATV Category**, Team Gensus, IITB (Dec'23)